



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Retail Sales Performance Analysis For Business Decision Making

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:



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3. Problem Statement

Retail businesses generate large volumes of transactional data on a daily basis, including information related to product sales, customer purchases, pricing, and regional performance. However, this raw data does not provide meaningful insights unless it is properly analyzed. Organizations often face challenges in identifying sales trends, understanding product demand, and evaluating regional performance. This lack of structured analysis leads to inefficient decision-making in areas such as inventory management, product strategy, and revenue growth.

Additionally, while tools like Microsoft Excel are widely available, many users are unable to fully utilize their capabilities for data analysis. There is a need for a simple and structured approach that enables users to transform raw data into meaningful business insights. This project addresses these challenges by applying data cleaning, analysis, visualization, and basic AI-assisted techniques to support better decision-making in the retail domain.

4. Objectives

- To collect and explore the raw dataset, understand its structure, fields, and data quality issues.
- To clean and preprocess the data by removing duplicates, handling missing values, correcting formats, and standardizing fields.
- To apply domain-relevant analysis using the assigned tool (Excel formulas and pivot tables, Power BI measures, or AI prompt techniques).
- To build an interactive dashboard, structured report, or AI-generated output that communicates insights clearly to stakeholders.
- To integrate AI tools (Microsoft Copilot or AI Chatbot) to automate, enhance, or accelerate the analysis and content generation process.
- To document the complete project process, upload all files to GitHub, and record a demonstration video for submission.

5. Methodology / Implementation Plan

#	Phase	Description
1	Data Collection & Exploration	The dataset was obtained from an online source and explored to understand columns, structure, and data types

2	Data Cleaning & Preprocessing	Removed duplicates, handled missing values, and corrected formatting issues
3	Analysis & AI Integration	Used pivot tables and Excel functions to derive insights; AI tools were used for summarization
4	Visualization / Output	Created charts and a dashboard to represent sales trends and performance
5	Documentation & Submission	Prepared report, synopsis, PPT, and uploaded files to GitHub

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010 08:26	7.65	17850	United Kingdom

Fig 1: Raw dataset

InvoiceNo	Product Code	Description	Units Sold	Date of Transaction	Time of Transaction	UnitPrice	CustomerID	Sales Region	Transaction Type	Total transaction value
536365	85123A	White Hanging Heart T-Light Holder	6	01-12-2010	08:26	₹ 2.55	17850	United Kingdom	Sale	₹ 15.30
536365	71053	White Metal Lantern	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84406B	Cream Cupid Hearts Coat Hanger	8	01-12-2010	08:26	₹ 2.75	17850	United Kingdom	Sale	₹ 22.00
536365	84029G	Knitted Union Flag Hot Water Bottle	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84029E	Red Woolly Hottie White Heart.	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	22752	Set 7 Babushka Nesting Boxes	2	01-12-2010	08:26	₹ 7.65	17850	United Kingdom	Sale	₹ 15.30

Fig 2: Cleaned Dataset

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)

- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset used for the project is already available.
- Online Retail Dataset sourced from Kaggle: <https://www.kaggle.com/datasets>

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection and exploration	<i>Week 1</i>
2	Data cleaning and preprocessing	<i>Week 2</i>
3	Data analysis and insight generation	<i>Week 3</i>
4	Dashboard creation and visualization	<i>Week 4</i>
5	Report writing and submission	<i>Week 5</i>

7. Expected Outcomes / Conclusion

- A cleaned and structured dataset ready for analysis
- A dashboard showing sales trends, regional performance, and product insights
- Identification of key business insights to support decision-making
- Use of AI tools to enhance documentation and analysis
- A complete project including report, PPT, and GitHub repository

8. References

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: <https://www.linkedin.com/learning>

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